ETC3250/5250 Introduction to Machine Learning

Week 7: Explainable artificial intelligence (XAI)

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Overview

We will cover:

- Global explainability - what is the affect of this variable on the response?
  - Variable Importance
  - Partial dependence profiles
- Local explainability - Why was this observation predicted like that?
  - LIME
  - Counterfactuals
  - Anchors
  - Shapley values

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Black-Box Machine learning

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Black-Box Machine learning

Interpretable Models

- Logistic regression, LDA and (a single) Decision Tree are interpretable machine learning models
- It is easy to see why any one observation was predicted the way it was (local)
- and understand the effect any one variable has on the model's prediction (global)
- In logistic regression and LDA this comes from looking at the model coefficients
- In Decision trees we can look at the tree structure (as long as it is not too complex)

Un-Interpretable Models

- Random Forests and Boosting models with many trees,
- and Neural Networks with many hidden nodes or layers
- are not interpretable

- It is not obvious why any observation was predicted how it was
- or what the affect of any variable was on the response variable

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Tradeoff

- We need complex models to fit complex dataset
- But the model is not being used by us (data scientists/machine learners)
- It is used by some decision maker to help make a decision
  - They are unlikely to blindly trust you and use your model
  - Saying it has a 0.95 out of sample AUC is unlikely to convince them
  - They will only trust your model if they can explain what it is doing

Local vs Global Explainability

- *Local*: Any question related to how a particular prediction for a particular observation was made;
- *Why for this particular patient did the model recommend not to treat?*

- *Global*: How the model behaves overall, or for a particular target feature of subset.
- *How does certain increase in GDP would affect the labor rate?*

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Global explainability

How does certain increase in GDP would affect the labor rate?

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Variable importance (1/3)

Remember:

- In Logistic regression and LDA (when we standardise the predictors) the larger the model coefficient the more important that variable for the prediction
- In DTs and RFs we had the gini index
- Importance from permutation

However

- Effect of collinearity

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Variable importance (2/3)

Model coefficients on standardised data

```
parsnip model object

Call:
lda(species ~ ., data = data, prior = ~c(1/3, 1/3, 1/3))

Prior probabilities of groups:
   Adelie Chinstrap    Gentoo
     0.33      0.33      0.33

Group means:
             bl    bd    fl    bm
Adelie    -0.94  0.61 -0.78 -0.62
Chinstrap  0.90  0.64 -0.36 -0.58
Gentoo     0.66 -1.10  1.16  1.09

Coefficients of linear discriminants:
```

`bl` and `bd` are the most important variables

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Variable importance (3/3)

When predictors are strongly linearly associated, interpreting coefficients purely on magnitude can be incorrect.

Original

```
     LD1    LD2
bl -0.24 -2.319
bd  2.04  0.172
fl -1.22  0.062
bm -1.18  1.257
```

Correlated variables

```
       LD1    LD2
bl    1.80 -1.921
bl2  -1.57 -0.407
bd2   0.49  1.548
bd   -2.54 -1.375
fl    1.21  0.052
bm    1.21  1.270
```

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Permutation variable importance (1/2)

For trained model $\hat{f}$, which depends on data $\mathbf{X}$ to predict response y, with loss function $L(y, \hat{f})$ (e.g. misclassification rate, error),

1. Estimate $L(y, \hat{f})$ on the data, L^{orig}.
2. For each variable $j \in 1, \ldots, p$,
   - Generate data matrix $\mathbf{X}^{perm}$ by permuting variable j. This breaks the association between variable j and observed y.
   - Compute the $L(y, \hat{f})$ on the permuted data, L^{perm}.
   - Compare L^{orig} and L^{perm}, e.g. $|L^{orig} - L^{perm}|$
3. Most important variables have larger values.

Permutation variable importance (2/2)

Random forests have this baked into the model fitting (using the out-of-bag cases).
Generally, should be conducted on the test set 9out of sample).

```
# Using DALEX with tidymo
# https://www.tmwr.org/ex
# https://ema.drwhy.ai/fe
vip_features <- colnames(

vip_train <-
  p_std |>
  select(all_of(vip_featu

explainer_lda <-
  explain_tidymodels(
    lda_fit,
    data = vip_train,
    y = p_std$species,
    verbose = FALSE
```

```
  )
vip_lda <- model_parts(ex
                       B=
```

Data with additional correlated variables

- Variables with correlation still can affect results.
- Variables can mask the importance of others.

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Partial dependence profiles (1/2)

Partial dependence profiles show how the model prediction changes across different values of an explanatory variable.

Computes

$$\hat{p}(Y = k \mid X_j = x) =$$

$$\frac{1}{n}\sum_{i=1}^{n} \hat{p}(Y = k \mid X_j = x, X_{-j} = x_{i-j})$$

Where $x_{i-j} = x(x_{i1}, \ldots, x_{i,j-1}, x_{i,j+1}, \ldots, x_{ip})$

```
# With DALEX
pdp_lda <- model_profile(
            explainer_lda,
            N=100)
```

Shows what the model sees.

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Partial dependence profiles (2/2)

What do we see?

PDP suggests LDA sees

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Local explainability

Why for this particular patient did the model recommend not to treat?

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Why should I know about local explainers?

If you deploy a complex model, you **may** need to be able to explain any decision made from it.

If the decisions affect people or organisations, they might be challenged in court. You as the analyst may be expected to justify the decision, that it was made fairly, without bias, and based on specific measurements collected for the model.

The goal

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Global vs Local

- Linear models are global models, the affect of one variable is the same whatever the affect of the other variables

$$P(Y = 1 \mid X) = \frac{1}{1 + \exp(-\beta_0 - \beta_1 x_1 - \beta_2 x_2 - \cdots - \beta_p x_p)}$$

- When the model is non-linear, the effect of different variables changes locally
- Take for example a decision tree
  - Iteratively split on a single variable
  - If first split on x_1 and then on both sides split on x_2, the effect of x_2 will be different depending on what x_1 was
- The same can be said in NN's
  - Take the ReLU activation ReLU$(x) = \max(x, 0)$

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- One variable may causes a neuron to fire with some combination of the other variables, but with other combinations it may not

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Linear vs non-linear separation

Mark a point where x1 is most important in distinguishing the classes. Around about (0, 0), if $x_2 = 0$ then $x_1 < 0$ is blue and $x_1 > 0$ is red (also aroud $(0.75, 0.5)$)

Mark a point where x2 is most important in distinguishing the classes. Around about (0, -0.5), if $x_1 = 0$ then $x_2 > -0.5$ is blue and $x_2 < -0.5$ is red

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Selected points to use for illustration

Which variable is most important?

obs	expect
1	x2
2	x1
3	x1, x2
4	x1, x2
5	x1, x2
6	x1

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LIME

Fit a linear regression in the local neighbourhood of observation of interest.

1. Introduce distance $d(\cdot, \cdot)$ and calculate the distance between all features x_i and $x_{i'}$.
2. For each observation $i \in 1, \ldots, n$,
   - Fit a linear model (logistic or linear regression) estimating $\beta^{(i)}$ maximising the log-likelihood where the contribution of each observation i' is weighted by $w_{i'} = \exp(-d(x_i, x_{i'}))$
3. If the data is standardised the magnitude of the $\beta_j^{(i)}$'s tells us how important variable j was to the prediction of variance i

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LIME

Weighted according to closeness to (2)

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LIME

Weighted according to closeness to (1)

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LIME

```r
library(DALEXtra)
library(lime)
w_rf <- randomForest(cl~., data=w)
w_rf_exp <- DALEX::explain(model = w_rf,
                           data = w[, 1:2],
                           y = w$cl == "A")
model_type.dalex_explainer <-
  DALEXtra::model_type.dalex_explainer
predict_model.dalex_explainer <-
  DALEXtra::predict_model.dalex_explainer
w_lime <- predict_surrogate(
  explainer = w_rf_exp,
              new_observation = w_new[,1:2],
              n_features = 2,
              n_permutations = 1000,
              type = "lime")
```

$P(Y = 1 \mid X) = \frac{1}{1+\exp(-\beta_0 - x_1\beta_1 - x_2\beta_2)}$

Class 1 = ‘A’ = Blue

```
#A tibble: 6 × 4
  case  model_intercept    x1      x2
```

```
  <chr>           <dbl>  <dbl>   <dbl>
1 1               0.579 -0.402  0.0939
2 2               0.569 -0.196 -0.0544
3 3               0.423  0.199  0.348
4 4               0.636 -0.362 -0.425
5 5               0.403  0.340  0.361
6 6               0.578  0.310 -0.514
```

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Counterfactuals

Find the closest observation (counterfactual) that has the different class.

What values of the variables would you need to change to change the observation of interest into the counterfactual.

```
library(iml)
# devtools::install_github("dandls/counterfactuals")
library(counterfactuals)
predictor_rf = iml::Predictor$new(w_rf,
                                  type = "prob")
# predictor_rf$predict(w_new[1,])
w_classif <- counterfactuals::NICEClassif$new(
  predictor_rf)

w_new_cf <- w_new
w_new_cf$cl <- ifelse(w_new[,3]=="A",
                           "B", "A")
for (i in 1:nrow(w_new)) {
  w_cf = w_classif$find_counterfactuals(
    x_interest = w_new[i,],
    desired_class = w_new_cf[i,3],
                 desired_prob = c(0.5, 1)
  )
  w_new_cf[i, 1] <- w_cf$data$x1
  w_new_cf[i, 2] <- w_cf$data$x2
}
```

```
   x1o   x2o clo      x1    x2 cl
1  0.5  0.25   A -0.5000 -0.31  B
2  0.0  0.00   B -0.0057  0.00  A
```

```
3  0.2 -0.50   B  0.1358 -0.50  A
4 -0.8  0.80   A -0.1785  0.51  B
5  0.8 -0.80   B  0.1358 -0.52  A
6  0.8  0.50   A  0.8249  0.50  B
```

Note: If case is misclassified, the desired class needs to be the true class.

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Anchors

How far can you extend from the value of the observation in each direction and still have all observations be the same class.
How big is a rectangle that contains each observation and only other observations of the same class

Note: No working R package to calculate these.

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Shapley values

Intuitively, a variable is important if the model's predictions change a lot if I take that variable away

Let $g(X_1, \ldots, X_p)$ be the machine learning model
Then variable j is important for observation i if
$|g(x_{i1}, \ldots, x_{ip})$
$-g(x_{i1}, \ldots, x_{ij-1}, x_{ij+1}, \ldots, x_{ip})|$
is large

However, if variable j is correlated with another variable then it can be important and this distance can look small

Therefore we need to consider the model with an without variable j, alongside all other combinations of the variables
e.g.
$|g(x_{ij} \cap x_{iS}) - g(x_{iS})|$
where S is all subsets of $\{1, \ldots, p\} \setminus j$

Averaged across all sets S is a Shapley value

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Shapley values

A Shapley value is computed from the change in prediction when all combinations of presence or absence of other variables.
In the computation, for each combination, the prediction is computed by substituting absent variables with their average value.

```
library(kernelshap)
library(shapviz)
w_explain <- kernelshap(
    w_rf,
    w_new[,1:2],
    w[,1:2],
    verbose = FALSE
  )
```

```
    x1    x2 cl shapAx1 shapAx2
1 -0.5 -0.25  A   0.358    0.15
2  0.0  0.00  B  -0.236   -0.25
3  0.2 -0.50  B  -0.164   -0.32
4 -0.8  0.80  A   0.255    0.26
```

```
5  0.8 -0.80  B  -0.215   -0.27
6  0.8  0.50  A  -0.059    0.57
```

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Summary

Which variable is most important?

obs	expect	LIME	CF	SHAP
1	x2	x1	x2	x1
2	x1	x1	x1	x1, x2
3	x1 x2	x2	x1	x2
4	x1, x2	x1, x2	x1, x2	x1, x2
5	x1, x2	x2	x1, x2	x1, x2
6	x1	x2	x1	x2

They don't all agree.

You need good visualisation of the model in the data space to fully digest the importance of the variables.

NOTE: We can use magnitude when interpreting the local explainers because we used standardised data. The interpretations are more

complicated otherwise.

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Example: penguins (1/2)

Compute SHAP values for the neural network model

```r
library(keras3)
#p_nn_model <- load_model_tf("../data/penguins_cnn")
p_nn_model <- keras3::load_model("../data/penguins_nn.h5"
p_nn_model

# Explanations
# https://www.r-bloggers.com/2022/08/kernel-shap/
library(kernelshap)
library(shapviz)
p_explain <- kernelshap(
    p_nn_model,
    p_train_x,
    bg_X = p_train_x,
    verbose = FALSE
  )
p_exp_sv <- shapviz(p_explain)
save(p_exp_sv, file="../data/p_exp_sv.rda")
```

Highlight SHAP values for a misclassified Gentoo penguin

Note: the SHAP values for misclassified penguins are much lower than values for all other penguins on `bm` and are higher than `bl`. i.e. `bm` is less important and `bl` is more important for those observations' prediction

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Example: penguins (2/2)

Weights from hidden layer

```
       [,1]  [,2]
[1,]  0.63  1.36
[2,]  0.17  0.08
[3,] -0.11 -0.30
[4,] -0.77 -0.24
```

```
[1] "bl" "bd" "fl" "bm"
```

Model uses mostly `bl` and `bm`.

Note: this analysis used the training set because this Gentoo penguin was misclassified as an Adelie in the training set.

```
          p_train_pred_cat
           Adelie Chinstrap Gentoo
Adelie        96         3      1
Chinstrap      2        43      0
Gentoo        38         0     44
```

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Other Uses

- XAI methods also gives **YOU** the data scientist/machine learner a way to understand your model
- Gives you a method do understand why your model is or is not working
- Which in turn allows you to think about how to improve it

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Next: Support vector machines and nearest neighbours

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